

# RHYZEN FASTFOOD POS SYSTEM

## MVC Core Point of Sale & Inventory Management System Documentation

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### 1. Project Overview

The **RHYZEN FASTFOOD POS SYSTEM** is a modern Point of Sale and Inventory Management System developed using **ASP.NET Core MVC Architecture**.

The system is designed specifically for fast food businesses to simplify and automate daily operations such as:

- Order Processing
- Sales Transactions
- Inventory Monitoring
- Product Management
- Receipt Generation
- Sales Reporting
- Administrative Management

The project focuses on:

- Real-world business workflow
  - Clean and responsive user interface
  - Efficient transaction handling
  - Scalable MVC architecture
  - Database-driven operations
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### 2. System Objectives

#### Main Objectives

- Improve transaction speed
- Automate inventory monitoring
- Reduce manual business operations
- Provide accurate sales tracking
- Enhance overall management efficiency

#### Specific Objectives

- Manage food products and categories
  - Handle customer orders efficiently
  - Track inventory stock levels
  - Generate transaction receipts
  - Secure administrator access
  - Store and manage business data using SQL Server
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### **3. Technologies Used**

#### **Frontend**

- HTML5
- CSS3
- JavaScript
- Bootstrap

#### **Backend**

- ASP.NET Core MVC
- C#

#### **Database**

- Microsoft SQL Server

#### **Development Tools**

- Visual Studio
  - SQL Server Management Studio (SSMS)
  - GitHub
  - Git
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### **4. System Architecture**

#### **MVC Core Architecture**

Model → Handles database and business logic

View → Handles user interface and frontend display

Controller → Processes requests and controls application flow

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## System Flow

User Input



Controller



Model



SQL Server Database



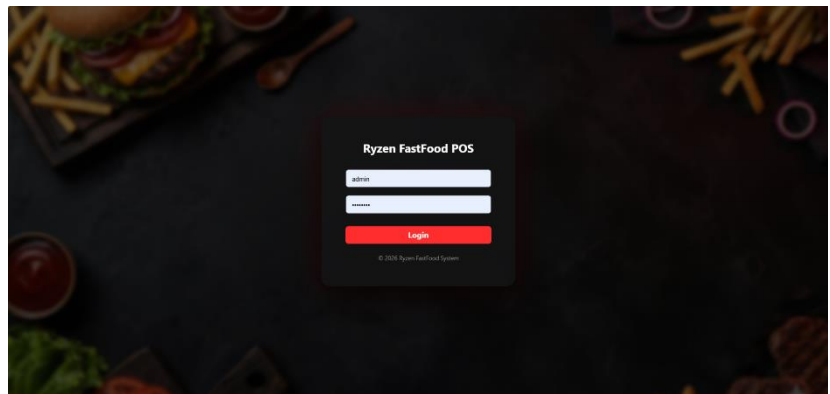
View Response to User

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## 5. Features of the System

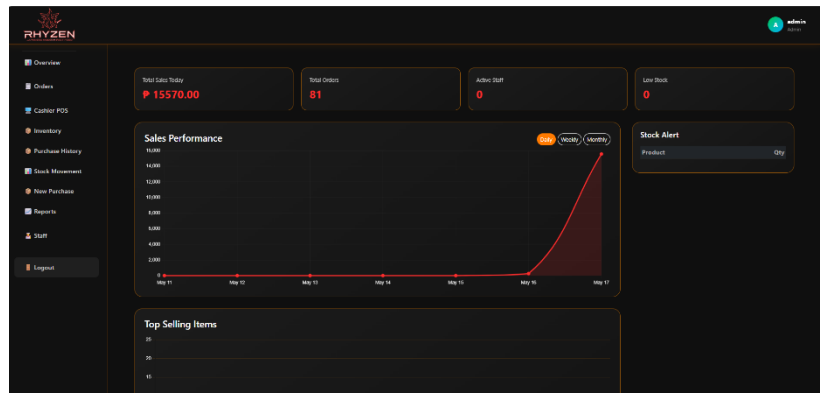
### 5.1 User Authentication

- Secure login system
- Administrator authentication
- Protected management access



### 5.2 Dashboard

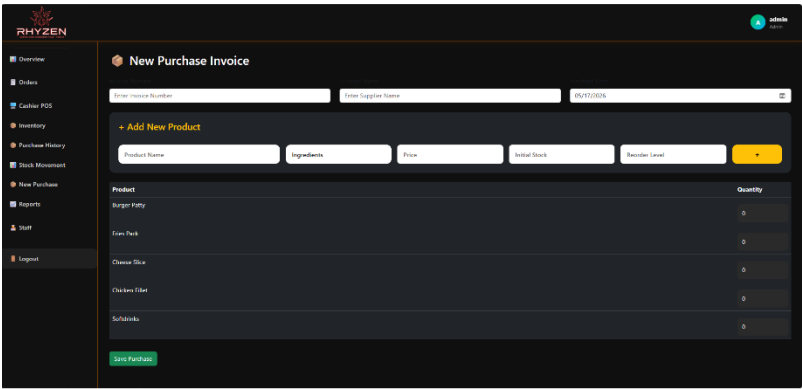
- Daily sales overview
- Inventory status
- Transaction summaries
- Business monitoring interface
- Complete Invoice tracking
- Product data analytics



| Invoice #         | Customer | Date         | Total    | Payment | Status    |
|-------------------|----------|--------------|----------|---------|-----------|
| ORD-177888117487  | Walk-in  | May 17, 2024 | P 180.00 | Cash    | Completed |
| ORD-177888452255  | Walk-in  | May 17, 2024 | P 180.00 | Cash    | Completed |
| ORD-177888452870  | Walk-in  | May 17, 2024 | P 180.00 | Cash    | Completed |
| ORD-1778884591799 | Walk-in  | May 17, 2024 | P 270.00 | Cash    | Completed |
| ORD-1778884591803 | Walk-in  | May 17, 2024 | P 270.00 | Cash    | Completed |
| ORD-1778884591823 | Walk-in  | May 17, 2024 | P 270.00 | Cash    | Completed |
| ORD-1778884591828 | Walk-in  | May 17, 2024 | P 270.00 | Cash    | Completed |
| ORD-1778884591838 | Walk-in  | May 17, 2024 | P 270.00 | Cash    | Completed |

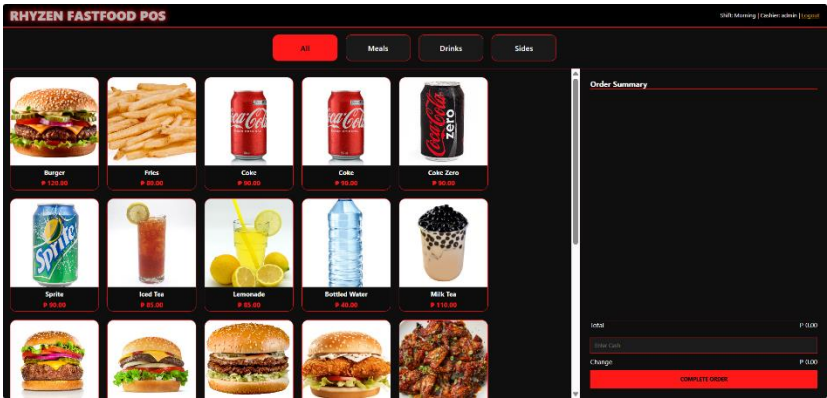
### 5.3 Product Management

- Add food products
- Edit product information
- Delete products
- Manage food categories
- Update pricing and stock



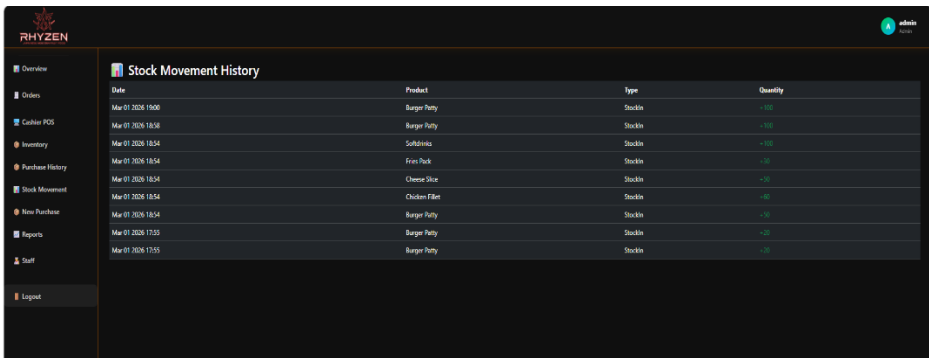
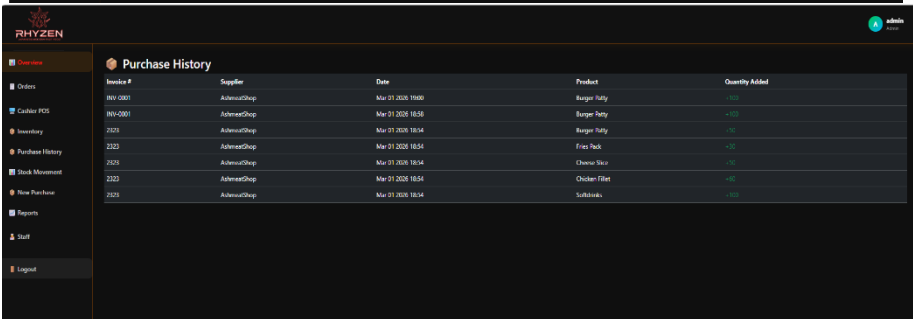
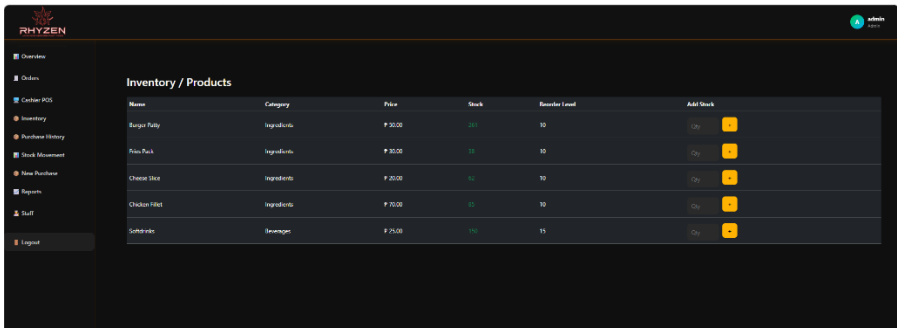
### 5.4 POS Transaction Module

- Fast order processing
- Add-to-cart functionality
- Quantity adjustment
- Automatic total computation
- Receipt generation



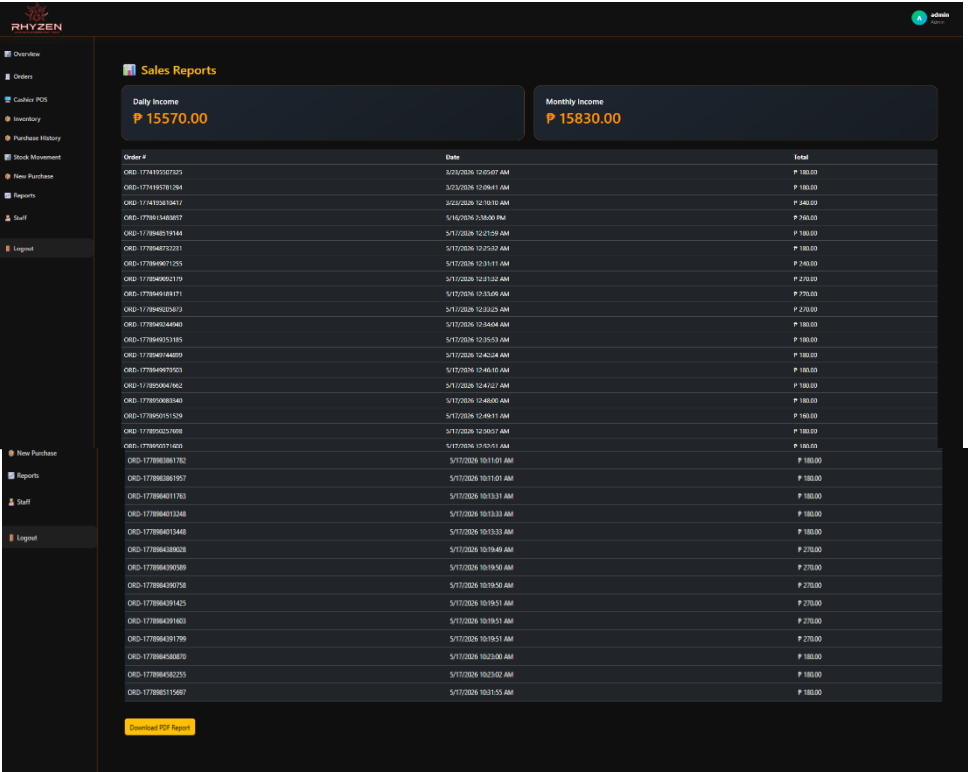
### 5.5 Inventory Management

- Real-time stock monitoring
- Automatic stock deduction
- Inventory tracking system
- Low stock management



5.6 Sales Reporting

- Transaction history
- Sales summaries
- Revenue tracking
- Order monitoring



6. Database Structure

Main Database Tables

